1.4 Abstract Algebra I: Solutions

These are solutions to problems.md.

Warm-up

1.

(a) **No.** Subtraction is not associative: $(1-1)-1=-1$ but $1-(1-1)=1$.
(b) **Yes.** Multiplication is associative and closed on nonzero rationals, 1 is the identity, and $1/q$ is the inverse of q.
(c) **Yes.** The set is closed under multiplication: it is $\langle i\rangle$, cyclic of order 4. It contains 1, and the inverses are $1^{-1}=1$, $(-1)^{-1}=-1$ and $i^{-1}=-i$.
(d) **No.** It is not closed: $\det(AB)=4$. It also does not contain the identity, since $\det I=1$.
(e) **A group iff n is prime.** If n is prime, $\mathbb{Z}_n$ is a field, so the nonzero classes form a group under multiplication. If $n=ab$ with $1<a,b<n$, then $a\cdot b\equiv 0$, which is not in the set, so the set is not closed. (For $n=2$ the set is $\{1\}$, the trivial group.)

2.

Cycle type	Example	Count	Order	Sign
1^4 (identity)	e	1	1	$+$
$2\,1^2$ (transposition)	$(1\,2)$	$\binom{4}{2}=6$	2	$-$
$3\,1$ (3-cycle)	$(1\,2\,3)$	$4\cdot 2=8$	3	$+$
4 (4-cycle)	$(1\,2\,3\,4)$	$3!=6$	4	$-$
2^2 (double transposition)	$(1\,2)(3\,4)$	3	2	$+$

The counts: there are 4 ways to choose the three points of a 3-cycle and 2 cyclic orders, giving 8. A 4-cycle can be written starting at 1, leaving 3! orders. There are 3 ways to split $\{1,2,3,4\}$ into two pairs. Total: $1+6+8+6+3=24$. The even permutations number $1+8+3=12=|A_4|$.

3. $\mathbb{Z}_{12}$ is cyclic, so it has exactly one subgroup for each divisor d of 12, namely $\langle 12/d\rangle$:

- $d=1$: $\{0\}$
- $d=2$: $\langle 6\rangle=\{0,6\}$
- $d=3$: $\langle 4\rangle=\{0,4,8\}$
- $d=4$: $\langle 3\rangle=\{0,3,6,9\}$
- $d=6$: $\langle 2\rangle=\{0,2,4,6,8,10\}$
- $d=12$: $\mathbb{Z}_{12}$

The generators are the k with $\gcd(k,12)=1$: $1,5,7,11$.

4. If $g^2=e$, then $g=g^{-1}$, for every g. So for any a,b: $ab=(ab)^{-1}=b^{-1}a^{-1}=ba$.

5. (a) If $\gcd(k,n)=1$, Bézout gives $ka+nb=1$ for some integers a,b, so $[k][a]=[1]$. Conversely, if $[k][a]=[1]$, then $ka-1=nb$ for some b, so any common divisor of k and n divides 1.

So every nonzero element of $\mathbb{Z}_n$ is a unit iff every k with $1\le k\le n-1$ is coprime to n, i.e. iff n is prime.

(b) $U(15)=\{1,2,4,7,8,11,13,14\}$, of order $\varphi(15)=8$. Orders, computed mod 15:

- $|1|=1$.
- $2,4,8,1$: $|2|=4$.
- $4^2=16=1$: $|4|=2$.
- $7^2=49=4$, so $|7|=4$.

- $8^2 = 64 = 4$, so $|8| = 4$.
- $11^2 = 121 = 1$: $|11| = 2$.
- $13^2 = 169 = 4$, so $|13| = 4$.
- $14^2 = 196 = 1$: $|14| = 2$.

No element has order 8, so $U(15)$ is **not cyclic**. It is abelian of order 8 with an element of order 4 and is not cyclic. Among $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$ and $\mathbb{Z}_2^3$, only $\mathbb{Z}_4 \times \mathbb{Z}_2$ fits. So $U(15) \cong \mathbb{Z}_2 \times \mathbb{Z}_4$. This also follows from CRT: $U(15) \cong U(3) \times U(5) \cong \mathbb{Z}_2 \times \mathbb{Z}_4$.

6. In a direct product, $|(a, b)| = \operatorname{lcm}(|a|, |b|)$.

(⇐) If $\gcd(m, n) = 1$, then $|(1, 1)| = \operatorname{lcm}(m, n) = mn$, which is the order of the group. So $(1, 1)$ generates.

(⇒) If $d = \gcd(m, n) > 1$, then every element has order dividing $\operatorname{lcm}(m, n) = mn/d < mn$. So no element generates.

Core

7. *Lagrange.* Let $H \leq G$ with G finite.

- The relation $a \sim b \iff a^{-1}b \in H$ is an equivalence relation. Reflexive: $e \in H$. Symmetric: inverses. Transitive: closure.
- Its classes are the left cosets aH, so the cosets partition G.
- Each coset has $|H|$ elements, because $h \mapsto ah$ is a bijection $H \to aH$ with inverse $x \mapsto a^{-1}x$.

So $|G| = [G : H] \cdot |H|$.

(a) If $|G| = p$ is prime and $g \neq e$, then $|\langle g \rangle|$ divides p and is greater than 1. So it equals p, and $G = \langle g \rangle$.

(b) $U(p) = \mathbb{Z}_p^\times$ has order $p - 1$. For $[a] \in U(p)$, $|[a]|$ divides $p - 1$, so $[a]^{p-1} = [1]$, i.e. $a^{p-1} \equiv 1 \pmod{p}$.

8. Let $K = \ker \varphi$. It is a subgroup: $\varphi(e) = e$, and if $a, b \in K$ then $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e$. It is normal: for $g \in G$ and $k \in K$, $\varphi(gkg^{-1}) = \varphi(g)\, e\, \varphi(g)^{-1} = e$.

First Isomorphism Theorem. Define $\bar{\varphi} : G/K \to \operatorname{im} \varphi$ by $\bar{\varphi}(gK) = \varphi(g)$.

- *Well defined and injective:* $aK = bK \iff a^{-1}b \in K \iff \varphi(a^{-1}b) = e \iff \varphi(a) = \varphi(b)$. The forward direction gives well-definedness, and the backward direction gives injectivity.
- *Homomorphism:* $\bar{\varphi}(aK\, bK) = \bar{\varphi}(abK) = \varphi(ab) = \varphi(a)\varphi(b)$.
- *Surjective onto* $\operatorname{im} \varphi$*:* immediate from the definition.

9. Let $[G : H] = 2$. The left cosets are H and $G \setminus H$, and so are the right cosets. If $g \in H$, then $gH = H = Hg$. If $g \notin H$, then $gH = G \setminus H = Hg$. So $gH = Hg$ for all g, which is equivalent to normality.

In S_3, let $H = \{e, (1\,2)\}$. Then $(1\,3)(1\,2)(1\,3)^{-1} = (3\,2) \notin H$, using the conjugation formula with $\sigma = (1\,3)$. So H is not normal. Its index is 3, which is consistent with the first part.

10. $\det : GL_n(\mathbb{R}) \to \mathbb{R}^\times$ is a homomorphism, since $\det(AB) = \det A \det B$. It is surjective: $\det \operatorname{diag}(t, 1, \ldots, 1) = t$. Its kernel is $SL_n(\mathbb{R})$, which is therefore normal by Problem 8. The First Isomorphism Theorem gives $GL_n(\mathbb{R})/SL_n(\mathbb{R}) \cong \mathbb{R}^\times$.

11.

(a) $\varphi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $(a, b) \mapsto a - b$, is a surjective homomorphism: $(n, 0) \mapsto n$. Its kernel is $\{(a, a)\} = \langle (1, 1) \rangle$. So the quotient is $\cong \mathbb{Z}$.

(b) $\varphi : \mathbb{R} \to S^1 = \{z \in \mathbb{C} : |z| = 1\}$, $t \mapsto e^{2\pi i t}$, is a homomorphism from $(\mathbb{R}, +)$ to $(S^1, \times)$. It is surjective, and its kernel is $\mathbb{Z}$. So $\mathbb{R}/\mathbb{Z} \cong S^1$, the circle group.

(c) $\varphi : \mathbb{Z}_4 \times \mathbb{Z}_2 \to \mathbb{Z}_2 \times \mathbb{Z}_2$, $(a, b) \mapsto (a \bmod 2,\ b)$. This is well defined because $2 \mid 4$, and it is a surjective homomorphism. Its kernel is $\{(0,0), (2,0)\} = \langle (2,0) \rangle$. So the quotient is $\cong \mathbb{Z}_2 \times \mathbb{Z}_2$. It is not $\mathbb{Z}_4$, because every element is killed by 2.

12. (a) The rotation group T of a regular tetrahedron acts on the 4 vertices.

The action is transitive. Let v, w be distinct vertices and let u be a third vertex. Rotation by $\pm 120°$ about the axis through u and the centre of the opposite face cyclically permutes the other three vertices, which include v and w. One of these two rotations takes v to w.

The stabiliser of a vertex v consists of rotations about the axis through v and the centre of the opposite face: angles $0°$, $120°$ and $240°$. That is 3 elements.

So $|T| = 4 \cdot 3 = 12$. In fact $T \cong A_4$.

(b) The rotation group of the cube acts transitively on the 6 faces. Rotations by $90°$ about the three face axes take any face to an adjacent face, and two such steps reach the opposite face. The stabiliser of a face is the 4 rotations about the axis through its centre. So the group has order $6 \cdot 4 = 24$.

13. *Lemma.* If $G/Z(G)$ is cyclic, then G is abelian.

Proof. Let $G/Z = \langle gZ \rangle$, where $Z = Z(G)$. Then every element of G has the form $g^i z$ with $z \in Z$. For $x = g^i z_1$ and $y = g^j z_2$:

$$xy = g^{i+j} z_1 z_2 = yx,$$

because z_1, z_2 are central and powers of g commute with each other.

Main proof. Let $|G| = p^2$. Then $Z(G) \neq \{e\}$, so $|Z(G)| \in \{p, p^2\}$. If $|Z(G)| = p$, then $|G/Z(G)| = p$, so $G/Z(G)$ is cyclic, and the lemma says G is abelian. But then $Z(G) = G$ has order p^2, a contradiction. Hence $Z(G) = G$, and G is abelian. (In fact $G \cong \mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.)

14. (a) Let $\tau = \sigma(a_1\ \ldots\ a_k)\sigma^{-1}$.

- $\tau(\sigma(a_i)) = \sigma(a_1\ \ldots\ a_k)(a_i) = \sigma(a_{i+1})$, with indices mod k.
- If $x \notin \{\sigma(a_1), \ldots, \sigma(a_k)\}$, then $\sigma^{-1}(x) \notin \{a_i\}$ is fixed by the cycle, so $\tau(x) = x$.

So $\tau = (\sigma(a_1)\ \ldots\ \sigma(a_k))$. Applying this to each cycle of a disjoint-cycle decomposition shows that conjugation preserves cycle type. Conversely, any two permutations of the same cycle type are conjugate: choose σ to send one cycle decomposition to the other, entry by entry.

(b) The class of $(1\,2)(3\,4)$ is all double transpositions: $\{(1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\}$, of size 3. By orbit–stabiliser, $|C_{S_4}((1\,2)(3\,4))| = 24/3 = 8$. It is a dihedral group of order 8 and a Sylow 2-subgroup of S_4.

(c) Suppose $H \leq A_4$ with $|H| = 6$. Then $[A_4 : H] = 2$, so $H \trianglelefteq A_4$, and A_4/H has order 2. So for every $g \in A_4$, $(gH)^2 = H$, i.e. $g^2 \in H$.

If σ is a 3-cycle, then $\sigma = \sigma^4 = (\sigma^2)^2 \in H$. So all eight 3-cycles lie in H. That is impossible, since $|H| = 6$.

15. (a) $15 = 3 \cdot 5$. n_5 divides 3 and $n_5 \equiv 1 \pmod 5$, so $n_5 = 1$. n_3 divides 5 and $n_3 \equiv 1 \pmod 3$, so $n_3 = 1$. Let P (order 3) and Q (order 5) be the Sylow subgroups. Both are normal.

- $P \cap Q = \{e\}$, by Lagrange.
- $|PQ| = 15$, so $PQ = G$.
- For $x \in P$ and $y \in Q$, the commutator $xyx^{-1}y^{-1}$ lies in P (because Q is normal, $yx^{-1}y^{-1} \in P$) and in Q (because P is normal, $xyx^{-1} \in Q$). So it is e, and $xy = yx$.

Hence $G \cong P \times Q \cong \mathbb{Z}_3 \times \mathbb{Z}_5 \cong \mathbb{Z}_{15}$, by Problem 6.

(b) $|G| = 20 = 2^2 \cdot 5$. n_5 divides 4 and $n_5 \equiv 1 \pmod 5$, so $n_5 = 1$. The unique Sylow 5-subgroup is normal of order 5, which is neither 1 nor 20. So G is not simple.

16. $|G| = 30 = 2 \cdot 3 \cdot 5$.

- n_5 divides 6 and $n_5 \equiv 1 \pmod 5$, so $n_5 \in \{1, 6\}$.
- n_3 divides 10 and $n_3 \equiv 1 \pmod 3$, so $n_3 \in \{1, 10\}$.

Suppose G is simple. Then $n_5 = 6$ and $n_3 = 10$. Distinct subgroups of prime order meet only in e. So there are $6 \cdot 4 = 24$ elements of order 5 and $10 \cdot 2 = 20$ elements of order 3. That is $44 > 30$ elements, a contradiction. So $n_3 = 1$ or $n_5 = 1$, and G has a proper nontrivial normal subgroup.

17. (a) *Prime.* R/I is nonzero iff $I \neq R$. In R/I, $(a + I)(b + I) = 0 + I$ iff $ab \in I$. So "no zero divisors in R/I" says exactly: $ab \in I \Rightarrow a \in I$ or $b \in I$.

Maximal. Suppose I is maximal and $a \notin I$. Then $I + (a)$ is an ideal strictly containing I, so it equals R. Write $1 = i + ra$ with $i \in I$. Then $(r + I)(a + I) = 1 + I$, so every nonzero element of R/I is invertible. Also $I \neq R$, so $1 \neq 0$ in R/I, and R/I is a field.

Conversely, suppose R/I is a field and $J \supsetneq I$ is an ideal. Pick $a \in J \setminus I$. There is r with $ra - 1 \in I \subseteq J$. Since $ra \in J$, we get $1 \in J$, so $J = R$. Hence I is maximal.

(b) Evaluation at 0, $\mathbb{Z}[x] \to \mathbb{Z}$, $f \mapsto f(0)$, is a surjective ring homomorphism. Its kernel is (x): $f(0) = 0$ iff the constant term is 0 iff $x \mid f$. So $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$. This is an integral domain but not a field, so (x) is prime but not maximal. Explicitly, $(x) \subsetneq (2, x) \subsetneq \mathbb{Z}[x]$.

18. Suppose $(2, x) = (f)$.

- Since $2 \in (f)$, f divides 2 in $\mathbb{Z}[x]$. Degrees add, so f is a constant, and $f \in \{\pm 1, \pm 2\}$.
- If $f = \pm 2$, then $x \in (2)$, which is false: the coefficient 1 is odd.
- If $f = \pm 1$, then $1 \in (2, x)$, so $1 = 2g(x) + xh(x)$. Comparing constant terms gives $1 = 2g(0)$, which is impossible in $\mathbb{Z}$.

So $(2, x)$ is not principal, and $\mathbb{Z}[x]$ is not a PID.

The map $\mathbb{Z}[x] \to \mathbb{F}_2$, $f \mapsto f(0)$ mod 2, is a surjective ring homomorphism. Its kernel is $\{f : f(0) \text{ even}\}$. Any such f can be written $f = f(0) + xg$, which lies in $(2, x)$; and $(2, x)$ is clearly inside the kernel. So $\mathbb{Z}[x]/(2, x) \cong \mathbb{F}_2$, and $(2, x)$ is maximal.

19. Use $N(a + bi) = a^2 + b^2$. It is multiplicative, and the units are the elements of norm 1: $\pm 1, \pm i$. *Criterion:* if $N(z)$ is a prime number, then z is irreducible. If $z = uv$, then $N(u)N(v)$ is prime, so one of u, v has norm 1 and is a unit.

(a)

- $2 = (1 + i)(1 - i) = -i(1 + i)^2$.
- $5 = (2 + i)(2 - i)$.
- $13 = (3 + 2i)(3 - 2i)$.

All the factors have prime norm (2, 5 or 13), so they are irreducible. Note that $1 - i = -i(1 + i)$ is an associate of $1 + i$.

(b) Suppose $3 = uv$ with neither factor a unit. Then $N(u)N(v) = 9$ and $N(u), N(v) > 1$, so $N(u) = 3$. But $a^2 + b^2 = 3$ has no integer solutions, since squares are $0, 1, 4, \ldots$. So 3 is irreducible. (In fact a rational prime p stays prime in $\mathbb{Z}[i]$ iff $p \equiv 3 \pmod 4$.)

(c) Divide, rounding the exact quotient to the nearest Gaussian integer.

- $\frac{11+3i}{1+8i} = \frac{(11+3i)(1-8i)}{65} = \frac{35-85i}{65} \approx 0.54 - 1.31i$. Take $q = 1 - i$. Then $(1 - i)(1 + 8i) = 9 + 7i$, so $11 + 3i = (1 - i)(1 + 8i) + (2 - 4i)$.

- $\frac{1+8i}{2-4i} = \frac{(1+8i)(2+4i)}{20} = \frac{-30+20i}{20} = -1.5 + i$. Take $q = -2 + i$. Then $(-2+i)(2-4i) = 10i$, so $1 + 8i = (-2+i)(2-4i) + (1-2i)$, with $N(1-2i) = 5 < 20$.
- $2 - 4i = 2(1-2i) + 0$.

So $\gcd(11+3i, 1+8i) = 1-2i$, up to units. Check: $N(11+3i) = 130$ and $N(1+8i) = 65$ are both divisible by $5 = N(1-2i)$. Also $11 + 3i = (1-2i)(1+5i)$ and $1 + 8i = (1-2i)(-3+2i)$.

20. (a) $x^3 - 2$ is Eisenstein at $p = 2$: $2 \mid 0, 0, -2$, $2^2 \nmid -2$, and $2 \nmid 1$. So it is irreducible over $\mathbb{Q}$. Alternatively: it has degree 3, and by the rational root test the only candidate roots are $\pm 1, \pm 2$, none of which is a root.

(b) The map $f(x) \mapsto f(x+1)$ is a ring automorphism of $\mathbb{Q}[x]$, so f is irreducible iff $f(x+1)$ is. Now

$$(x+1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2,$$

which is Eisenstein at 2. So $x^4 + 1$ is irreducible over $\mathbb{Q}$.

(c) Over $\mathbb{F}_2$, $(x+1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1 = x^4 + 1$. So $x^4 + 1 = (x+1)^4$.

21. $f(0) = 1$ and $f(1) = 1 + 1 + 1 = 1$ in $\mathbb{F}_2$. So f has no roots, and since it has degree 2, it is irreducible. Hence $K = \mathbb{F}_2[x]/(f)$ is a field with $2^2 = 4$ elements: 0, 1, α and $\beta = \alpha + 1$, where $\alpha = x + (f)$ satisfies $\alpha^2 = \alpha + 1$ (since $-1 = 1$).

Addition (characteristic 2, so $y + y = 0$):

$+$	0	1	α	β
0	0	1	α	β
1	1	0	β	α
α	α	β	0	1
β	β	α	1	0

Multiplication:

- $\alpha^2 = \beta$.
- $\alpha\beta = \alpha^2 + \alpha = 2\alpha + 1 = 1$.
- $\beta^2 = \alpha^2 + 2\alpha + 1 = \alpha^2 + 1 = \alpha$.

$\times$	1	α	β
1	1	α	β
α	α	β	1
β	β	1	α

$K^\times = \{1, \alpha, \beta\}$, with $\alpha^1 = \alpha$, $\alpha^2 = \beta$ and $\alpha^3 = \alpha\beta = 1$. So $K^\times = \langle\alpha\rangle \cong \mathbb{Z}_3$.

22. (a) Let $(p) \neq 0$ be prime, and suppose $(p) \subseteq (m)$. Then $p = mr$ for some r. Since p is a prime element, $p \mid m$ or $p \mid r$.

- If $p \mid m$, then $(m) \subseteq (p)$, so $(m) = (p)$.
- If $p \mid r$, write $r = ps$. Then $p = mps$, so $p(1 - ms) = 0$. Since $p \neq 0$ and R is a domain, $ms = 1$, so m is a unit and $(m) = R$.

So (p) is maximal.

(b) $F[x]$ is a PID, since it is Euclidean.

(⇐) Let f be irreducible. Then $(f) \neq F[x]$, because f is not a unit. If $(f) \subseteq (g)$, then $f = gh$, and irreducibility forces either g to be a unit, so $(g) = F[x]$, or h to be a unit, so $(g) = (f)$. Every ideal is principal, so (f) is maximal, and $F[x]/(f)$ is a field by Problem 17.

(⇒) If $f = gh$ with $1 \leq \deg g, \deg h < \deg f$, then $g, h \notin (f)$, because nonzero multiples of f have degree at least $\deg f$. But $(g+(f))(h+(f)) = 0$. So $F[x]/(f)$ has zero divisors and is not a field.

Challenge

23. Let $\lambda : G \to \mathrm{Sym}(G)$, $\lambda_g(x) = gx$. This is an injective homomorphism (Cayley). Since $|G|$ is even, Cauchy's theorem gives $t \in G$ of order 2.

The permutation λ_t has no fixed points, because $tx = x$ forces $t = e$. Also $\lambda_t^2 = \lambda_{t^2} = \mathrm{id}$. So λ_t is a product of $|G|/2 = m$ disjoint transpositions $(x\ \ tx)$. Since m is odd, λ_t is an odd permutation.

So $\mathrm{sgn} \circ \lambda : G \to \{\pm 1\}$ is a surjective homomorphism. Its kernel N is normal of index 2, so $|N| = m$. If $m > 1$, then N is neither trivial nor all of G, so G is not simple.

24. *Splitting criterion.* Let $x \in A_n$ and let $c = |\text{class of } x \text{ in } S_n| = n!/|C_{S_n}(x)|$. We have $C_{A_n}(x) = C_{S_n}(x) \cap A_n$, which is either all of $C_{S_n}(x)$ or a subgroup of index 2 in it.

- If $C_{S_n}(x) \subseteq A_n$, the A_n-class of x has size $(n!/2)/|C_{S_n}(x)| = c/2$. The S_n-class splits into two A_n-classes of size $c/2$, which are interchanged by conjugating by any odd permutation.
- Otherwise the A_n-class has size $(n!/2)/(|C_{S_n}(x)|/2) = c$, so it is the whole S_n-class.

The classes of A_5. The even cycle types in S_5 are 1^5, $3\,1^2$, $2^2\,1$ and 5.

- **3-cycles:** $\binom{5}{3} \cdot 2 = 20$ in S_5, so the centraliser has order $120/20 = 6$. It contains $(4\,5)$, which commutes with $(1\,2\,3)$ and is odd. So there is no splitting: one class of size **20**.
- **Double transpositions:** $5 \cdot 3 = 15$ in S_5, so the centraliser has order 8. It contains the odd element $(1\,2)$, which commutes with $(1\,2)(3\,4)$. One class of size **15**.
- **5-cycles:** $4! = 24$ in S_5, so the centraliser has order 5. It is $\langle (1\,2\,3\,4\,5) \rangle \subseteq A_5$. The class splits into **12 + 12**.
- **Identity: 1**.

Check: $1 + 20 + 15 + 12 + 12 = 60$.

Simplicity. A normal subgroup $N \trianglelefteq A_5$ is a union of conjugacy classes, contains $\{e\}$, and has order dividing 60. The possible sums $1 + (\text{a sub-multiset of } \{20, 15, 12, 12\})$ are

$$1, 13, 16, 21, 25, 28, 33, 36, 40, 45, 48, 60.$$

The only divisors of 60 in this list are 1 and 60. So $N = \{e\}$ or $N = A_5$, and A_5 is simple.

25. (a) $N(z) = z\bar{z}$, where $\bar{z}$ is complex conjugation, which preserves R. So $N(zw) = N(z)N(w)$. If u is a unit, then $N(u)N(u^{-1}) = N(1) = 1$, so $N(u) = 1$. Now $a^2 + 5b^2 = 1$ forces $b = 0$ and $a = \pm 1$. So the units are ± 1.

(b) Note that $a^2 + 5b^2$ never equals 2 or 3: if $b \neq 0$ it is at least 5, and if $b = 0$ it is a perfect square.

- $N(2) = 4$. A factorisation into non-units would need two factors of norm 2, which is impossible.
- $N(3) = 9$. It would need factors of norm 3, which is impossible.
- $N(1 \pm \sqrt{-5}) = 6$. It would need factors of norms 2 and 3, which is impossible.

So all four elements are irreducible.

(c) $6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$. The only units are ± 1, and $1 \pm \sqrt{-5} \neq \pm 2$, so 2 is not associate to either $1 \pm \sqrt{-5}$. These are therefore two essentially different factorisations into irreducibles, and R is not a UFD.

2 is not prime: $2 \mid (1+\sqrt{-5})(1-\sqrt{-5}) = 6$, but $2 \nmid 1 \pm \sqrt{-5}$, because $\frac{1 \pm \sqrt{-5}}{2} \notin R$. In a UFD every irreducible is prime, so this is another way to see that R is not a UFD.